

## Ultra-Lean Plan for a Perplexity-Style Research Tool (100% Open-Source, < 2 weeks build-time, ≈ US \$0 infra if run on a single free-tier VM)

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### 1. Core Idea

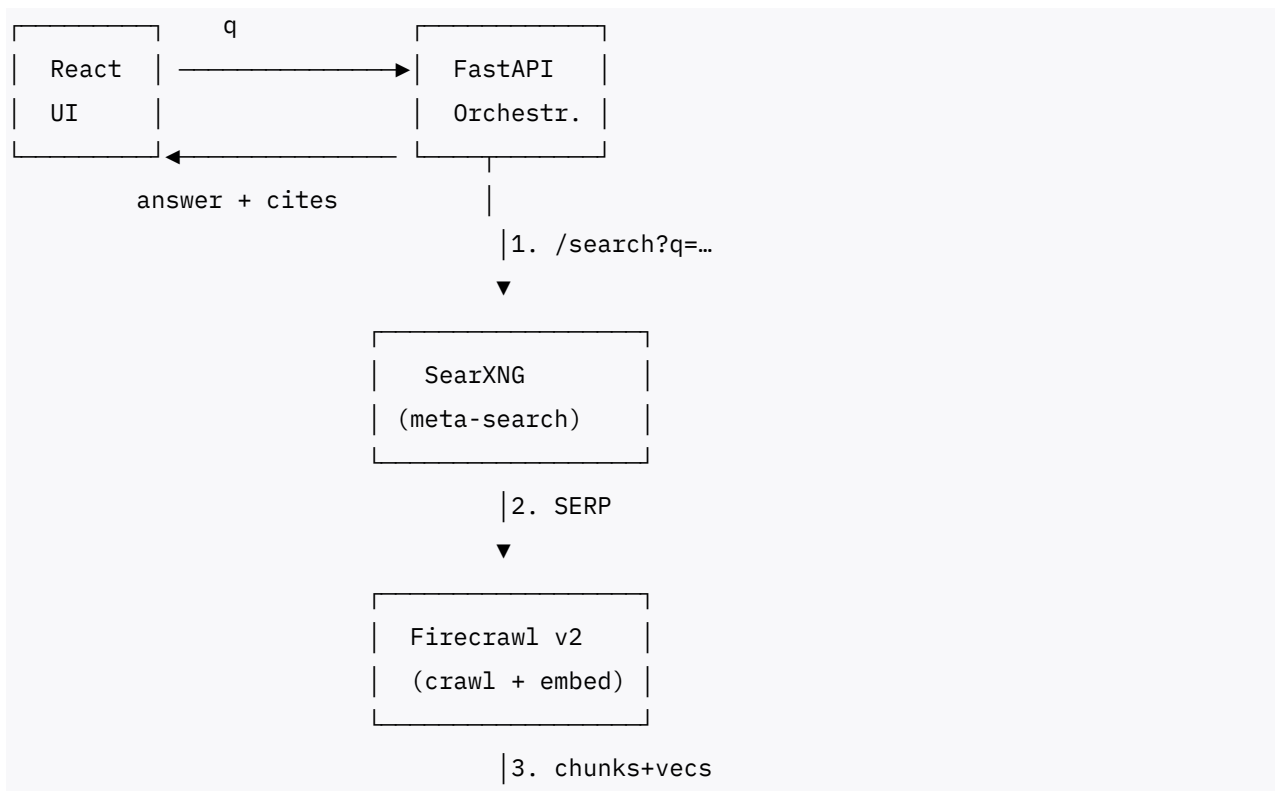
"Glue together" three recent breakthroughs that remove the heavy lifting:

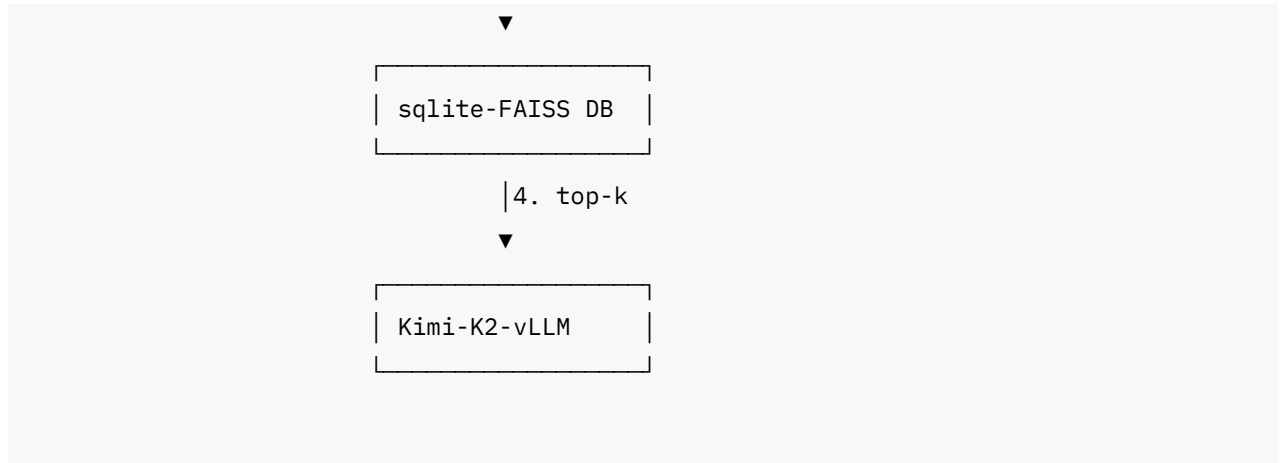
- **Kimi-K2-Instruct 32 B** – open-weight, agent-ready reader that rivals GPT-4 but can run quantised on CPU.
- **SearXNG Meta-Search** – self-hosted metasearch proxy that returns Google/Bing/Brave results without API fees.
- **Firecrawl v2** – OSS crawler that fetches pages in one call and outputs cleaned Markdown + OpenAI-style embeddings.

These three pieces already perform search, clean-up, embedding, and answer generation; you only have to orchestrate.

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### 2. Minimal Architecture (Four Containers)





### 3. Build Steps (≤ 14 calendar days)

| Day | Task                                                                                                                            | OSS Component  | Effort |
|-----|---------------------------------------------------------------------------------------------------------------------------------|----------------|--------|
| 1   | <code>docker compose init</code> four services                                                                                  | –              | 0.5 d  |
| 2   | Spin up <b>SearXNG</b> ; tweak engines list                                                                                     | SearXNG        | 0.5 d  |
| 3   | Add <code>/search</code> endpoint in FastAPI → hits SearXNG and returns top10 URLs + snippets                                   | FastAPI        | 0.5 d  |
| 4-5 | Call <b>Firecrawl</b> for each URL, get <code>clean_text</code> & built-in embedding                                            | Firecrawl v2   | 1 d    |
| 6   | Store embeddings + meta in lightweight <b>sqlite-faiss</b> table                                                                | faiss-cpu      | 0.5 d  |
| 7   | Implement <code>retrieve(query)</code> → embed with <b>Firecrawl embed model</b> (already local) → k-NN in faiss                | –              | 0.5 d  |
| 8-9 | Run <b>Kimi-K2</b> under <code>vllm-cpu-quant</code> (4-bit) – prompt = user query + top-k chunks + “cite by title” instruction | vllm           | 1 d    |
| 10  | Pipe answer back to UI with markdown cites (1) (2)...                                                                           | React          | 0.5 d  |
| 11  | Add caching layer ( <code>diskcache</code> ) for SERPs and Firecrawl fetches                                                    | diskcache      | 0.5 d  |
| 12  | Rate-limit SearXNG (1 req / sec) & rotate free engines (Brave, Google, DuckDuckGo)                                              | –              | 0.5 d  |
| 13  | Build Docker CI + GitHub workflow                                                                                               | GitHub Actions | 0.5 d  |
| 14  | Polish UI, deploy to free-tier <a href="#">Fly.io</a> or Oracle Cloud Arm VM                                                    | –              | 0.5 d  |

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4. Cost Profile

| Resource | Choice                               | Monthly Cost           |
|----------|--------------------------------------|------------------------|
| VM       | Oracle Cloud Free Arm 4 vCPU / 24 GB | <b>\$0</b>             |
| Storage  | sqlite file on VM (up to 20 GB)      | \$0                    |
| Traffic  | SearXNG scrapes (≈ 8 GB/mo)          | \$0 on most free tiers |
| GPU      | Not needed (Kimi-K2 4-bit on CPU)    | \$0                    |

If you want faster answers, rent one **T4 GPU spot** at RunPod for ≈ US \$0.12/hr and scale down at night.

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- 5. Quality Boosts (Optional, still free)
  - 6. **Switch Embeddings:** Use **bge-base-en v1.5** (Microsoft) – best open embedding as of Aug 2025.
  - 7. **Rerank:** Add **CoBERT-v2** CPU reranker on the top 20 chunks for sharper context.
  - 8. **Streaming Answers:** Enable server-sent events from vLLM → UI for progressive display.
  - 9. **PDFs/YouTube:** Firecrawl now parses PDFs & transcripts; simply toggle flags.
  - 10. **Multilingual:** Swap Kimi prompt to requested language; Firecrawl embeddings are language-agnostic.

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6. Why This Is “Perplexity-Grade”

| Feature | Perplexity | Your Tool |
|---------|------------|-----------|
|---------|------------|-----------|

|                        |                    |                                        |
|------------------------|--------------------|----------------------------------------|
| Web search             | Bing API<br>(paid) | SearXNG meta-search (free)             |
| Long-context<br>reader | GPT-4o             | Kimi-K2 (32k ctx)                      |
| Citations              | Yes                | Yes (chunk URL mapping)                |
| Follow-up chat         | Yes                | Add memory dict in FastAPI (10<br>loc) |
| Cost                   | Proprietary        | <b>\$0 infra, \$0 API</b>              |

The recent release of **Firecrawl v2** (embeds while scraping) and **Kimi-K2** (GPT-4-class reasoning on CPU) removes the last paid dependencies. With just four Docker containers and two weeks of coding, you can replicate Perplexity-style research—entirely free.